Sumit Gupta

Final Year M. Tech scholar, IIT Kanpur
Department of Civil Engineering

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ACADEMIC QUALIFICATIONS

Qualification	Institute	Year	Score
M.Tech – Civil Engineering (Hydraulics & Water Resources Engineering)	Indian Institute of Technology Kanpur	2024 – Present	CPI: 9.05 / 10
B.Tech – Civil Engineering	Institute of Engineering and Technology (AKTU), Lucknow	2019 – 2023	CGPA: 8.8 / 10
Class XII (CBSE)	Shri Ram Public School, Kanpur	2018	86.6%
Class X (CBSE)	Shri Ram Public School, Kanpur	2016	10 CGPA

TEACHING & ACADEMIC EXPERIENCE

- **Teaching Assistant – CE362 : Engineering Hydraulics (IIT Kanpur)** (Jan 2026 – Present)
  - Conducted tutorial classes for B.Tech students helping them understand subject clearly and improve their problem-solving skills
  - Assisted in preparing and evaluating assignments & ensuring proper evaluation
  - Solved student doubts through discussions
- **Teaching Assistant – CE610 : Advanced Hydrology (IIT Kanpur)** (Aug 2025 – Dec 2025)
  - Conducted tutorials and assisted students in understanding core engineering concepts
  - Guided students in assignments and problem-solving approaches and doubt discussion

INTERNSHIPS & CERTIFICATES

- **Central Public Works Department (CPWD) – Internship** ***(July 2022)***
  - Exposure to **construction practices, estimation, site management and preperation of Bill of Quantity (BOQ)** in public infrastructure projects.

RESEARCH WORK & ACADEMIC PROJECTS

- **M.Tech Thesis** | *Supervisor*: Dr. Tushar Apurv ***(June 2025 – Ongoing)***
  Title: **Spatio temporal analysis of Irrigation and Cropping Pattern Changes in Uttar Pradesh**
  - **Objective:** Identify discrepancies between groundwater- and rainfall-stressed zones & analyze changes in cropping and irrigation patterns and their impact on groundwater resources.
  - **Approach:** Used MODIS remote sensing data to analyze crop growth and evapotranspiration trends; examined temporal changes (2001–2023) in cropping patterns, irrigation sources, and groundwater levels; identified emerging water-stress in Uttar Pradesh.
- **B.Tech Project** | *Supervisor:* Prof. J.B Srivastava | *Co-Guide:* Abhishek Mishra ***(Completed – June 2023)***
  Title: **Reduction of Dynamic Lateral Earth Pressure on Retaining Walls using Waste Tyre as Backfill**
  - **Objective:** Analyze lateral earth pressure on backfill and reduce it using waste tyre chips.
  - **Approach:** Performed numerical analysis using OPTUM G2 to model retaining wall–backfill interaction, comparing conventional soil with waste tyre chip as backfill; results showed a reduction in lateral earth pressure with the use of tyre chips.
- **Optimal Water Allocation Modeling for Cauvery River Basin (KRS Reservoir, Karnataka–Tamil Nadu)**
  Course Project: CE718 Water Resources System Analysis ***(April 2025)***
  - Tools/Frameworks: LINGO (optimization modeling), Reservoir Simulation, Data Analysis.
- **IoT-Based Monitoring of Evapotranspiration Using Load Cell**
  Course Project: CE738A Hydrometry ***(Nov 2024)***
  - Tools/Frameworks: Arduino, Python, PostgreSQL, MQTT, EAGLE PCB, Data Analysis.
- **Analysis and Visualization of Gridded Rainfall Data over India**
  Course Project ***(March 2025)***
  - Applied multiprocessing for parallel rainfall analysis across grid points and generated spatial contour plots for variability assessment.
- **Watershed Delineation and LULC Analysis of Pandu River Basin**
  Course Project ***(April 2025)***
  - Tools/Skills: QGIS, Google Earth Engine, Hydrological Modeling, Spatial Analysis, Map Design.
- **Water Surface Profile Analysis using HEC-RAS**
  Course Project: CE613A ***(March 2025)***
  - Tools/Skills: HEC-RAS, Excel, Hydraulic Modeling, Flow Analysis.

TEACHING AREAS & SUBJECT EXPERTISE

1. Fluid Mechanics and Hydraulics
2. Engineering Hydrology
3. Open Channel Flow
4. Surveying
5. Environmental Engineering
6. Geotechnical and Foundation
7. Transportation
8. Solid mechanics
9. Construction Management

ADMINISTRATIVE EXPERIENCE

- **Department Placement Coordinator** , IIT Kanpur *(July 2025 – Ongoing)*
  - Proactively engaged with 200+ companies and HR managers, driving recruiter outreach, onboarding, and participation for campus hiring.
  - Served as the primary point of contact for recruiters, managing end-to-end placement operations including scheduling, communication, and coordination.
  - Streamlined placement processes and led resume reviews and student training initiatives, improving placement readiness and overall outcomes.
- **Student Guide, Institute Counselling Service**, IIT Kanpur *(April 2025 – Ongoing)*
  - Mentored and supported students in academic and personal development, facilitating smooth campus adaptation.
  - Guided students on academic planning, stress management, time manaagement and decision-making.
- **Teaching Assistant, Department Post Graduate Committee -** IIT Kanpur *(May 2025)*
  - Administered academic activities including assignment evaluation, doubt-solving sessions

INDUSTRY INTERACTION & PLACEMENT ACTIVITIES

- Coordinated placement drives, internship programs, and recruitment activities by serving as a liaison between students, faculty, and industry representatives.
- Organized career guidance sessions, resume-building workshops, mock interviews, and professional skill development programs to improve student employability.
- Assisted students in professional profile development, including resume review, interview preparation, and career planning.
- Supported outreach initiatives to attract new recruiters and expand internship and placement opportunities across diverse sectors.
- Coordinated end-to-end campus recruitment activities, including company outreach, recruiter engagement, visit planning, interview scheduling, stakeholder communication, and seamless execution of placement and internship drives involving students, faculty, HR teams, and corporate representatives.

POTENTIAL CONTRIBUTION TO THE INSTITUTION

- Assistant Professor in Civil Engineering
- Training and Placement Officer
- Student Career Development and Mentorship Coordinator
- Academic Administration and Department Coordination
- Research and Project Development

TECHNICAL SKILLS

- **Programming Languages:** Python
- **Software / Tools:** HEC-RAS | AutoCAD | QGIS

ACHIEVEMENTS

- Awarded **First Division with Distinction** in **B.Tech (Civil Engineering)** for academic excellence .
- Qualified **GATE - 2023** twice with a best rank of **AIR 4940 .**
- Secured **State Rank 1996** in **UPSEE - 2019** (Uttar Pradesh State Engineering Entrance Exam) .
- **JEE Mains Qualified (AIR - 48963, 2019)** .
- Essay Writing Competition (**1st rank**)– Bharat Vikas Parishad .

INTERESTS & ACTIVITIES

- **Professional & Personal Development:** Public Speaking, PPT Presentations, Entrepreneurship, Networking & Connection-Building.
- **Sports :** Cricket, Running, Badminton.